实验四 文件操作

实验内容

- 运用面向对象程序设计思想，基于Java文件管理和I/O框架，实现简易的文本文件内容搜索功能。
- 输入文件：项目根目录下的lab4文件夹以及其内部的所有文件（来自chinadaily.com.cn的新闻，部分文件在原新闻基础上有微小且不影响语义的改动）
- main方法的输出示例见下页（示例结果可能不完整，可能与真实结果不一致）

示例（存在精确匹配结果）

精确匹配

大小写不敏感

Please enter query keywords: **donald trump**

Exact match(es):

列出所有完整包含**query keywords**的句子以及句子所在的文件名，顺序自定

(1) Sentence "WASHINGTON - President-elect Donald Trump on Thursday tapped Susie Wiles, his campaign manager, to be White House chief of staff." in file "President-elect Trump taps Susie Wiles as White House chief of staff".

(2) Sentence "The tentative appointment came less than two days after Donald Trump beat his Democratic opponent and Vice President Kamala Harris, winning by landslide on issues including inflation, immigration and Middle East conflicts." in file "President-elect Trump taps Susie Wiles as White House chief of staff".

(3) Sentence "Donald Trump's election victory on Wednesday has raised concerns." in file "Let's hope new US president ends conflicts as promised".

Please enter query keywords: **chengdu**

如果一个句子出现多次**query keywords**，则只返回一个结果

Exact match(es):

(1) Sentence "Siemens Industrial Automation Products (Chengdu) Ltd in Chengdu, Sichuan province, as a world-class digitalized factory, has adopted a range of digital concepts and methods in practical applications to enhance production efficiency and quality." in file "AI applications driving enterprise productivity".

示例（不存在精确匹配结果）

如果没有精确匹配结果，但存在近似匹配结果，则返回近似匹配结果

query关键词的近似匹配词

```
Please enter query keywords: Doin George
Approximate match(es):
(1) Sentence ""COP30 is where findings must be translated into action," said Dion George,
South African minister of forestry, fisheries and the environment. "" in file "Action
urged ahead of UN climate meeting".
```

无结果返回

```
Please enter query keywords: iPhone17pro
No exact or approximate matches found.
```

输入*#00则退出

```
Please enter query keywords: *#00

Process finished with exit code 0
```

注意事项

- 如果在文档集合中找到了符合条件的精确匹配结果，则程序只输出精确匹配结果。精确匹配的查询的最小单元为keyword；keyword由空格或标点符号分隔（包括 ’）。具体而言，假设query keyword为”car”，且某个文件包含句子”I saw two cars.”, 则该句子不作为结果输出。
- 如果在文档集合中没有找到符合条件的结果，则程序改为近似匹配。大家需自行定义何为“近似”，并且把近似匹配的具体定义以注释形式写出来。此为开放式问题，无标准答案；合理即可。如果找到了近似结果，则输出结果。如果没有找到近似结果，则输出:
  - No exact or approximate matches found.
- 在实现过程中，大家在可根据个人理解进行合理的假设。

提交要求

- 提交代码文件：包含main方法的Main.java文件以及其他相关文件。
- 将源代码压缩打包发送至:
  - 邮件标题格式：实验4_姓名_学号
  - 压缩包文件名格式：实验4_姓名_学号
- 代码提交截止时间：2025年10月30日23:59:59
- 建议大家在实验课上完成所有内容